LAB03-INSTRUCTIONS



- **Problem-solving strategies**
- **Pseudocode and flowcharts**
- **Introduction to class and object (OOP)**
- **Let's use Git to track change** of your source codes as well during your development. When you are done with one exercises, you can commit the changes.
- After you are done, **create a report** containing pseudocode, flowchart diagram, your code screenshot and output along with some text explanation.

Recommendation Steps for Each Problem:

1. Understand the Problem: Identify inputs, processes, and outputs.
2. Write Pseudocode: Outline the logic step by step.
3. Draw Flowchart: Create a flowchart to visualize the logic.
4. Choose Data Structures: Let's create class and object in Object-oriented programming (OOP).
   Add more variables, constructor, and methods to your class.
5. Implement the Solution: Write your codes in Java programming language.

LAB03-PRACTICE1



Practice 1: Library Management System

Problem: Design a program to manage books in a library.

Input: For **N books**, store **Book ID**, **Title**, **Author**, and **Price**.

Process: Display details of all books, and calculate the **average price** of all books.

Output: Display all book details and the average price.

LAB03-PRACTICE2



Practice 2: Temperature Conversion

Problem: Write a program that reads temperatures of **N cities** in **Celsius** and converts them to **Fahrenheit.**

Input: Temperature of **N cities** in Celsius.

Process: Use the formula **F = (C × 9/5) + 32.**

Output: Display the temperatures in both Celsius and Fahrenheit for each city.

LAB03-PRACTICE3

Practice 3: Employee Salary Management

Problem: Store and display salary details of N employees

Input: For each employee, store **Employee ID**, **Name**, and **Basic Salary**.

Process: Calculate the **Gross Salary** using the formula:

- Gross Salary = Basic Salary + HRA (20% of Basic) + TCA (10% of Basic)

Output: Display all details, including the calculated gross salary.

- HRA (House Rent Allowance) is an allowance paid by employers to employees to cover their house rent expenses.
- TCA (Travel cost allowance) is an allowance paid to employees to the travel cost.

LAB03-PRACTICE4

Practice 4: Student Grades Calculation

Problem: Calculate and display student grades based on marks.

Input: For **N students**, read **Student ID**, **Name**, and marks of **5 subjects**.

Process: Calculate the **total marks**, **average**, and assign a grade:

- A: Average >= 90
- B: Average >= 75
- C: Average >= 50
- F: Average < 50

Output: Display all details, including total marks, average, and grad

LAB03-PRACTICE 5

Practice 5: Simple Banking System

Problem: Simulate a basic banking system

Input: For **N customers**, store **account number (14-digit)**, **name**, and **balance**.

Process: Allow the user to:

- Deposit money into an account.
- Withdraw money from an account (ensure sufficient balance).

Output: Display updated account details.